

Hello, micro:bit

TODAY YOU'LL MAKE

A micro:bit that smiles at you.

GET READY

- ☐ Laptop
- ☐ micro:bit
- ☐ USB cable
- ☐ Pencil

STEPS

- 1 Plug the micro:bit into the laptop with the USB cable.
- 2 Click **New Project**. Name it **Hello**.
- 3 Drag a **show icon** block inside **on start**. Pick the heart.
- 4 Click **Download**. Drag the file onto the **MICROBIT** drive.
- 5 Unplug the cable, plug it back in. Watch the lights.

IF IT BREAKS

Nothing happens after the download? Find the **MICROBIT** drive on the laptop and drag the file right onto it.

No MICROBIT drive? Try the other cable. Some cables only charge — they don't carry files.



SCAN ME

YOUR CHALLENGE

Swap the heart for your initial using **show string**. Then make it flash: icon, pause, clear screen.

DESIGN YOUR OWN 5×5 PICTURE**I DID IT!**

Date: _____

ONE THING I LEARNED

Two Buttons, Two Jobs

TODAY YOU'LL MAKE

A micro:bit that answers you.

GET READY

- ☐ Laptop
- ☐ micro:bit
- ☐ USB cable
- ☐ Pencil

STEPS

- 1 Click **New Project**. Name it **Buttons**.
- 2 Click **Input**. Drag out **on button A pressed**.
- 3 Drop **show string** inside it. Type **YES**.
- 4 Do the same for button B. Make it say **NO**.
- 5 Download it. Ask the micro:bit a question.

IF IT BREAKS

Only one button works? Each button needs its own block. Check one drop-down says A and the other says B.

Words go by too fast? show string scrolls once. Press the button again.



SCAN ME

YOUR CHALLENGE

Add **on button A+B pressed**. Make it show something only you would know.

WRITE YOUR TWO BUTTON MESSAGES



I DID IT!

Date: _____

ONE THING I LEARNED

Shake It

TODAY YOU'LL MAKE

A dice you shake instead of roll.

GET READY

- ☐ Laptop
- ☐ micro:bit
- ☐ USB cable
- ☐ Pencil

STEPS

- 1 Click **New Project**. Name it **Dice**.
- 2 Click **Input**. Drag out **on shake**.
- 3 Drop **show number** inside it.
- 4 Drag **pick random** from **Math** into the number slot. Make it 1 to 6.
- 5 Download it. Shake it twenty times.

IF IT BREAKS

It rolls when you don't want it to? You're bumping the table. Hold it flat and shake on purpose.

Keep getting 0? **pick random** starts at 0. Change the first box to 1.



SCAN ME

YOUR CHALLENGE

Make it roll two dice — two numbers with a pause between. Roll twenty times. Which total wins?

TALLY YOUR TWENTY ROLLS**I DID IT!**

Date: _____

ONE THING I LEARNED

Two micro:bits Talking

TODAY YOU'LL MAKE

A message sent across the room with no wires.

GET READY

- ☐ Laptop
- ☐ micro:bit ×2
- ☐ USB cable
- ☐ Battery pack
- ☐ Pencil

STEPS

- 1** Click **New Project**. Name it **Radio**.
- 2** Put **radio set group 1** inside **on start**.
- 3** On button A pressed, add **radio send number 1**.
- 4** Add **on radio received**. Put a **show icon** heart inside.
- 5** Download the **same** program to **both** micro:bits. Press A.

IF IT BREAKS

Nothing arrives? Both need the same group number and the same program. Download it twice.

Only one lights up? The sender doesn't show its own message. Add a **show icon** under the send block too.



SCAN ME

YOUR CHALLENGE

Walk to the far end of the house and press A. How far can you get before it stops? Mark the spot.

MAP HOW FAR THE SIGNAL REACHED

**I DID IT!**

Date: _____

ONE THING I LEARNED

Wear It

TODAY YOU'LL MAKE

A step counter you can take outside.

GET READY

- ☐ Laptop
- ☐ micro:bit
- ☐ USB cable
- ☐ Battery pack
- ☐ AAA batteries

STEPS

- 1 Click **New Project**. Name it **Steps**.
- 2 Click **Variables**. Make a variable called **steps**.
- 3 On shake, add **change steps by 1**.
- 4 On button A pressed, add **show number steps**.
- 5 Download it. Clip on the battery pack. Go for a walk.

IF IT BREAKS

Count way too high? One step can jiggle it twice. Walk exactly 10 steps and see what it says.

Goes blank when unplugged? Switch the battery pack on and push the plug all the way in.



SCAN ME

YOUR CHALLENGE

Count one lap of the garden. Then count Dad's lap. Whose legs take more steps — and why?

DRAW WHERE YOU'LL WEAR IT

**I DID IT!**

Date: _____

ONE THING I LEARNED
